"YOU MUST NOT ssh!" 🚨🤖

and Other Stories from Building Our SRE Agent

David Pech — Wrike

Who



David Pech

wrike

GOLDEN
Kubestronaut

The Golden Grail of SRE Agents

What KubeCon EU was about

- **Every other booth** had an "AI SRE"
- Same demo, same Slack screenshot
- Same promise: **no more on-call pages**

Reality Check

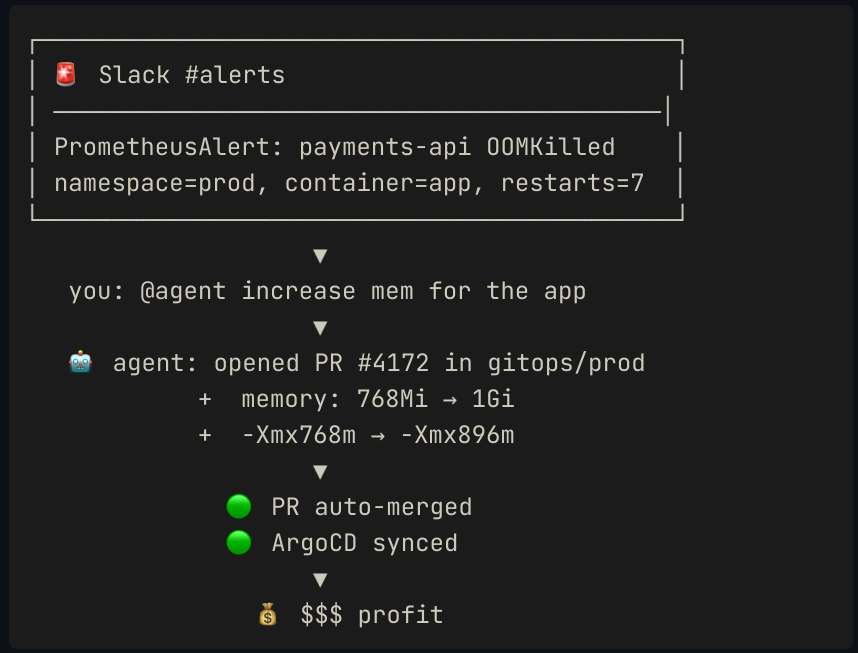
- 🏷️ Seeing 80% of booths for the last time
- 🧪 Just a few vendors with real production "read-write" experience
- 🔌 **MCP-everywhere** - just wrapping original API servers oftentimes

The pitch in one line

> Alert fires → agent investigates → agent fixes your precious production



Booth promise: the "$$$ profit" demo 💰



What this demo silently assumes

- 🧭 The agent already knows **which** GitOps repo, **which** overlay, **which** file
- 🪪 The agent has **write** credentials to a prod repo
- 🤝 The on-call person **trusts** a one-line Slack request
- 🧠 +25% memory is actually the right fix (not a leak, not a noisy neighbor)
- 🔁 No one cycles the agent into a feedback loop with itself

POC: Proof of Concept 🛠️



internal "Product" ← **the most important part**

🤷

Who is it for?

- We are not sure
- Ops + maybe: DevOps, Developers, Managers...

What should it do?

- We don't know
- Investigate, research, maybe: fix things? Do we really want that?

Covered use-cases:

- what is our setup of app Kafka cluster in QA
- do we serve domain X via Cloudflare?
- translate IP 10.1.2.3 to hostname

+ Slack `@integration` (now Claude Tag)

POC: Results ~January-March

- Python vibe-coded app on top of `Claude Agent SDK`
  - With MCPs to PagerDuty, VictoriaMetrics, Netbox, ...
- Tapped into Slack channels (both human and alerting)
- Bundled a lot of our repos
  - Wikis
  - Docs
  - GitOps repos
  - Terraform repos
  - ...
- Workflow: what to do (context)

Runtime	Credentials	Users	Tokens spent
locally + skills	`$HOME`	? SysOps Juniors (but we don't have those)	~$2k

POC: Mistakes

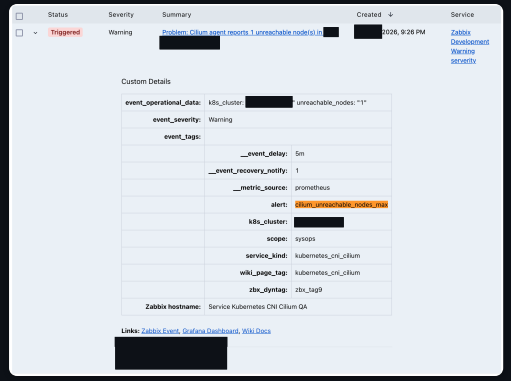
✨ AI magic will happen™ ✨

Main problem: too generic, not much added value.

Intended direction:

- ❌ Understand larger picture from multiple alerts at the same time
- ❌ Fix the issue in QA (note: PROD was never target)
- ❌ Keep session open
- ❌ Don't pollute with thinking
- ❌ Do not allow any tools except for `Read`, `Glob`...
- ❌ Semi-legacy Wiki exported into context

What Can You Get Without Context: Guess Work



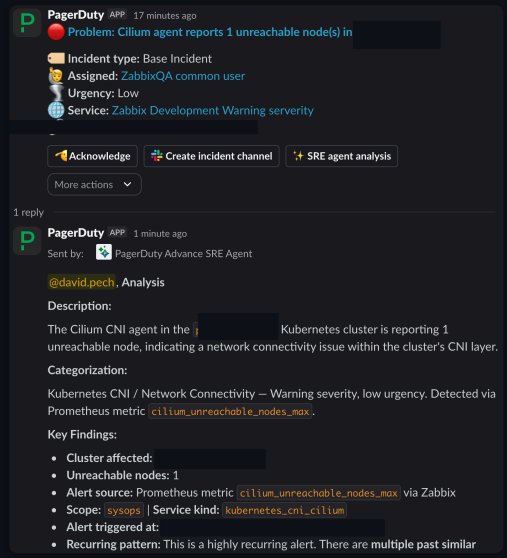
Status	Severity	Summary	Created ↓	Service
Triggered	Warning	Problem: Cilium agent reports 1 unreachable node(s) in [redacted]	[redacted] 2026, 9:26 PM	Zabbix Development Warning serverity

Custom Details

event_operational_data:	k8s_cluster: [redacted] unreachable_nodes: "1"
event_severity:	Warning
event_tags:	
__event_delay:	5m
__event_recovery_notify:	1
__metric_source:	prometheus
alert:	cilium_unreachable_nodes_max
k8s_cluster:	[redacted]
scope:	sysops
service_kind:	kubernetes_cni_cilium
wiki_page_tag:	kubernetes_cni_cilium
zbx_dyntag:	zbx_tag9
Zabbix hostname:	Service Kubernetes CNI Cilium QA

Links: Zabbix Event, Grafana Dashboard, Wiki Docs

[redacted]



PagerDuty APP 17 minutes ago

Problem: Cilium agent reports 1 unreachable node(s) in [redacted]

Incident type: Base Incident
Assigned: ZabbixQA common user
Urgency: Low
Service: Zabbix Development Warning serverity

Acknowledge | Create incident channel | SRE agent analysis | More actions

1 reply

PagerDuty APP 1 minute ago

Sent by: PagerDuty Advance SRE Agent

@david.pech, **Analysis**

Description:

The Cilium CNI agent in the [redacted] Kubernetes cluster is reporting 1 unreachable node, indicating a network connectivity issue within the cluster's CNI layer.

Categorization:

Kubernetes CNI / Network Connectivity — Warning severity, low urgency. Detected via Prometheus metric `cilium_unreachable_nodes_max`.

Key Findings:

- **Cluster affected:** [redacted]
- **Unreachable nodes:** 1
- **Alert source:** Prometheus metric `cilium_unreachable_nodes_max` via Zabbix
- **Scope:** `sysops` | **Service kind:** `kubernetes_cni_cilium`
- **Alert triggered at:** [redacted]
- **Recurring pattern:** This is a highly recurring alert. There are **multiple past similar**

Resume: Not bad given how little context it has.

Useful: absolutely not.

> @PagerDuty booth discussion on KubeCon NA 2025: "Can we add more context?" "No." (but it might have changed since...)

You Need to Provide More Context

Our approach:

- mimic the local Claude Code setup
- mono-repo style

Problem:

- CLAUDE.md (AGENTS.md) in each repo
- You should not provide out-of-date, legacy information **THIS IS BIG**

> 🔥 **Stale context = confident wrong answers.** Pruning beats hoarding.

MCP vs. tool use for Wiki

For some reason - MCP significantly less effective.

Medium sized wiki (1000 pages):

- grep and other basic agent tools are very effective

GitOps repos

Our approach:

- move from some proprietary wiki to Markdown
- we needed to recreate wiki from scratch

Typically 30 - 110 tool uses (Grep|Read|Glob) for a single prompt

~80MB of context, 8MB .tf, 3MB .md

Session vs. Separate each run

Note: Token caching: 5 mins

Our approach:

- each @mention is a new session

Trade-off:

- fewer tokens
- more time taken each run
- some details are lost

Surprisingly - not a problem

Example:

```
- Alert A.
- You: @ops-responder-agent - find the definition of the a
- Agent: Definition is ...
- You: @ops-responder-agent - based on the alert definitio
- Agent: (start from scratch)
```

Slack Context

Our original idea:

- we absolutely must share the context of all Slack alerting channels, so the agent is aware of the general situation and can leverage it

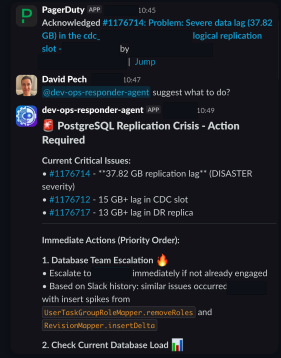
Workflow

Step 1: Locate the Prompt Message (MANDATORY)

Before doing ANYTHING else, locate the exact message that triggered this evaluation in `./slack/` history.

File Structure

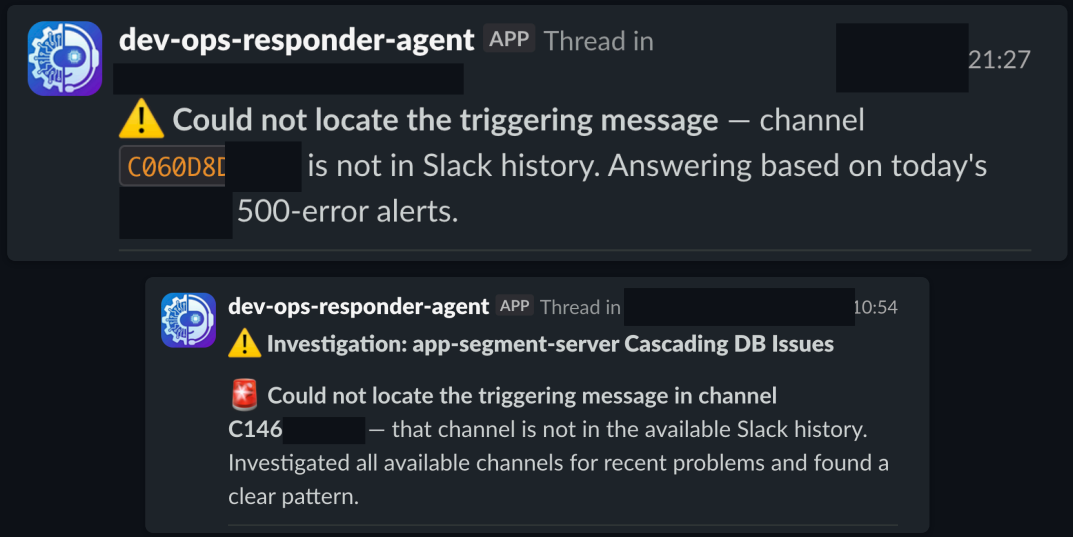
```
./slack/
  index.yaml                  # Channel list with names and message counts
  {channel_id}/
    recent.yaml               # Last 24h
    extended.yaml             # 1-7 days
```



POC	V1
Generate up-to-date Slack dump, caching + handling Slack	Replace with MCP called on-demand, saving 20s on each prompt (copying of the context each prompt)

Problem: Slack Context

- it uses tools like Grep|Sed - skips the context on purpose



POC	V1
No further support	CLAUDE.md's almost everywhere, guidance in the context

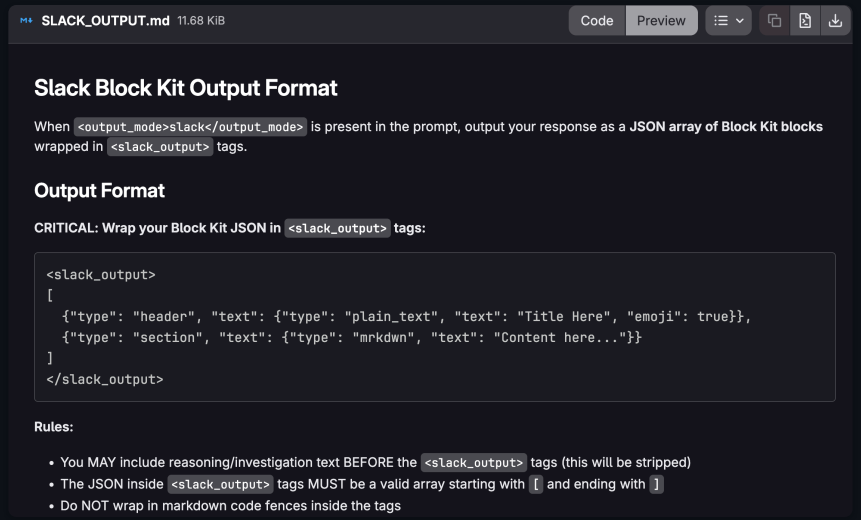
Problem: Context Or Code?

Slack Output, Slack Fetching

Hard to decide, even having both and experimenting between them.

Slack Output

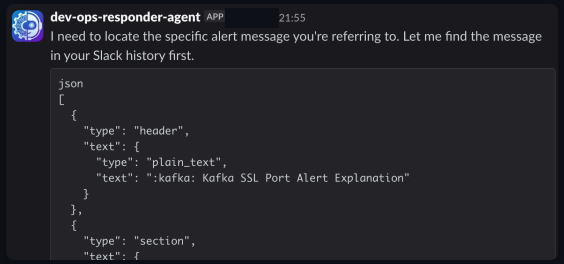
- no formatting
- have a script replace .md / something to `Block Kit`
- explain that in a context

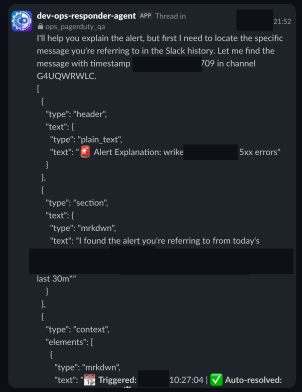


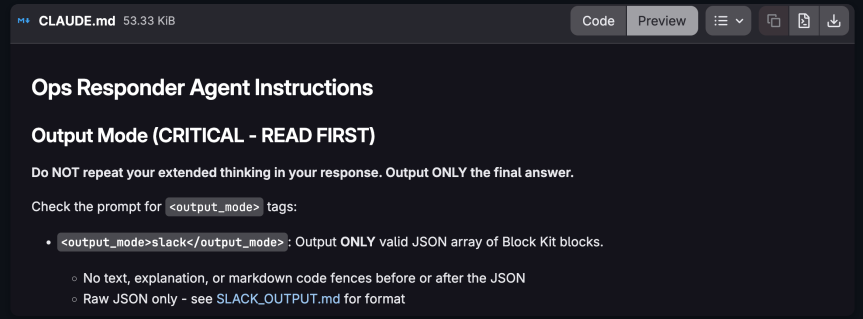
Problem: Output Gets Polluted

Thinking: off

"Extended thinking"







Problem: How to Get Result in a Reasonable Time



Highly model + thinking budget dependent.

It doesn’t work well. Number of steps was even worse.

Also - we can’t easily switch fast. vs in-depth answers.

Execution Timeout Awareness

You have a 6-minute (~360 second) timeout for each evaluation.

If your execution is forcibly terminated due to timeout, the user receives no response at all. To prevent this:

1. **Prioritize critical information first** - Answer the most important part of the question early in your response
2. **Be aware of time-consuming operations** - Multiple MCP queries, large file searches, and complex analysis chains take time
3. **If investigation is taking too long**, provide a partial answer with what you've found:
   - "⏳ *Investigation incomplete due to time constraints*"
   - Summarize findings so far
   - List what you were unable to check
   - Suggest the user can ask follow-up questions for specific areas
4. **Structure responses for early value** - Don't save your conclusions for the end. If you've identified the issue, state it immediately, then provide supporting details.
5. **When deep investigation is needed**, consider:
   - Answering the immediate question first
   - Noting "For a deeper analysis of X, please ask a follow-up question"

POC	V1
Opus	Sonnet

Problem: MCP on top of API

- MCP tools are not aligned to human prompting
- IDs vs. human-readable

Our approach:

- `/init`-like instruction for such MCPs

MCP Server usage

netbox

NetBox is the source of truth for IPAM and device inventory.

Hostname → IP

1. Find host: `netbox_get_objects(object_type="virtualization.virtualmachine", filters={"name__ic": "hostname"})` (or `dcim.device` for physical)
2. If `primary_ip4` is null, get interfaces: `netbox_get_objects(object_type="virtualization.vminterface", filters={"virtual_machine_id": <id>})`
3. Get IPs per interface: `netbox_get_objects(object_type="ipam.ipaddress", filters={"vminterface_id": <id>})`
   - **Query each interface separately** (one call per interface ID)
   - Do NOT use `vminterface_id__in` — NetBox silently ignores this unsupported suffix and returns ALL IPs (~8k+ results)

IP → Hostname

```
netbox_get_objects(object_type="ipam.ipaddress", filters={"address": "10.12.40.4"})
# Check assigned_object.virtual_machine.name in response
```

Problem: MCPs Expose Too Much Data

- Superadmin token will give you all the teams in your company
- You need only those that have category 'ops'

Either you spend tokens on each usage + time or it doesn’t work well.

Service Filtering

You MUST filter only the services (thus alerts) for 'Ops Team' (6YF), use 'service_ids' in API:

- BTM - Alert Prototype
- VAI - Alibaba cloud
- V1S - Analytics grafana
- R03 - Beacon Development
- MC - Beacon Production

Claude Code vs. Claude Agent SDK

Same thing, so they say...

Problem: 60s for MCP startup - fails 10% of the time

Our approach — retry the evaluation 3 times on this failure.

Problem: either use allowList for permission or callback function for "Bash" tool

Our approach: yes, we parse shell ourselves...

HolmesGPT (at the time): BB-gun

Problem: Is this Bash safe?

The agent wants to run tool Bash: `...`

(New feature - auto-mode classifier)

```
python vibe-coded-script-previously-written.py
```

...and try to get a security approval over here...

Our approach:

- Writing into the context folder is not allowed
- Only a very limited allowlist of commands, no piping `|`

Not great...

V1: What Actually Works ~April-August

Who is it for?

- SysOps
- some Engineering Managers

What should it do?

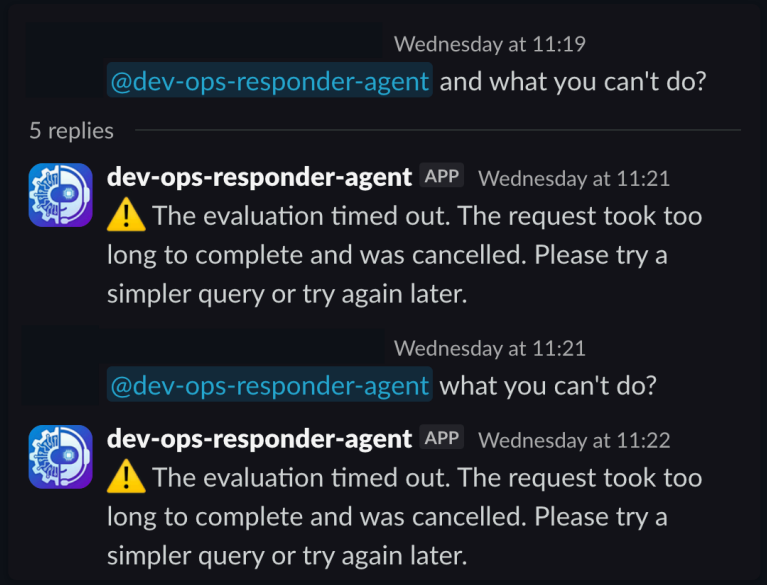
- automatically suggest actions
- react on user prompts

Covered use-cases:

- explain the request path for wrike.com/ (80% correct)
- is this alert recurring?
- are we vulnerable?
- ...

Runtime	Credentials	Users	Tokens spent
remote	shared read-only tokens	SysOps	~$5k

How People Interact With SRE Agent



Examples

dev-ops-responder-agent APP at 09:47

👋 Hi there! Thanks for making me available for testing. I'm an Ops Responder Agent designed to help with incident response and infrastructure monitoring.

What I can help with:
- Analyze current alerts and incidents from PagerDuty
- Look up infrastructure details in NetBox (IPs, hostnames)
- Query metrics from VictoriaMetrics
- Search historical incidents in Wrike

See more

How to use me effectively:
- Mention me with `@dev-ops-responder-agent` in channels or threads
- For thread context: reply to specific alerts/incidents I should analyze
- Use the `^` operator to reference recent messages above
- Be specific: mention service names, timeframes, or alert IDs when possible

Example questions:
- "What alerts are currently firing?"
- "Explain this PostgreSQL alert ^"
- "What's the IP of db-primary-01?"
- "Any similar incidents in the last 7 days?"

See more



David Pech Wednesday at 10:44

@dev-ops-responder-agent what can you do?

1 reply

dev-ops-responder-agent APP Wednesday at 10:45

👋 Here's what I can help with

I assist Wrike's ops team with incident response, infrastructure monitoring, metrics analysis, and documentation lookups. Here are some example questions:

🚨 **Incidents & Alerts**
- "What alerts are currently firing in PagerDuty?"
- "Who is currently on-call for the Kubernetes service?"

📈 **Metrics & Monitoring**
- "Check the JVM heap usage for srv- "

🌐 **Infrastructure Lookup**
- "Which host has IP 10.12. "

📚 **Documentation & Architecture**
- "Explain the Patroni failover process"

I can also search Wrike incident history, analyze Slack threads, look up alert configurations, and query live metrics from VictoriaMetrics.

💡 *Want more examples from a specific area? Just ask!*

General recon vs. research 🔍

PagerDuty APP 21:43
Unacknowledged #1181289: Disk space is low (used > 91%) on /var/lib/kafka: PROBLEM for kafka-1 by Zabbix Production High serverity | Jump

at 21:54
@dev-ops-responder-agent There are a lot of .log files in /var/lib/kafka/data, each file is 1 Gb in size. Is it safe just to remove some old ones?

dev-ops-responder-agent APP 21:56

🧯 **Do NOT manually delete Kafka log segment files**

❌ **No, it is NOT safe to manually remove `.log` files from `/var/lib/kafka/data`.**

Those `.log` files are **Kafka log segments** — they are **not** application logs. They contain the actual topic/partition data (messages). Deleting them will cause **data loss** and can **crash Kafka brokers**.

Why each file is ~1 GB:
Kafka splits each partition into segments. The default `log.segment.bytes` is typically set to 1 GB (1073741824 bytes). This is normal and expected behavior — each file represents one segment of a partition's commit log.

What happens if you delete them manually:
• ⚠️ **Data loss** — consumers that haven't read those messages will miss them permanently

---

✅ **The correct way to reclaim disk space:**

1. Adjust retention settings (preferred)
Let Kafka manage cleanup via broker or per-topic configuration:
• `log.retention.hours` — reduce retention period (e.g., 72 → 48 hours)
• `log.retention.bytes` — set max bytes per partition
• `log.cleanup.policy=delete` — ensures old segments are purged automatically

Show more

2. Delete entire topics (if topic is no longer needed)

```
kafka-topics.sh --delete --topic <topic>
```

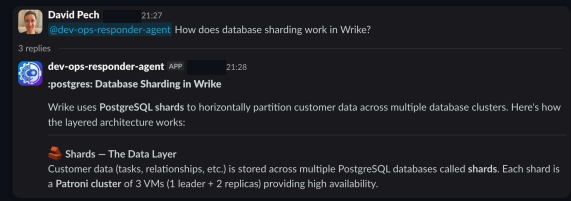
This cleanly removes all segments, indexes, and metadata.

3. Investigate what's consuming space
Before making changes, check which topics/partitions are largest:

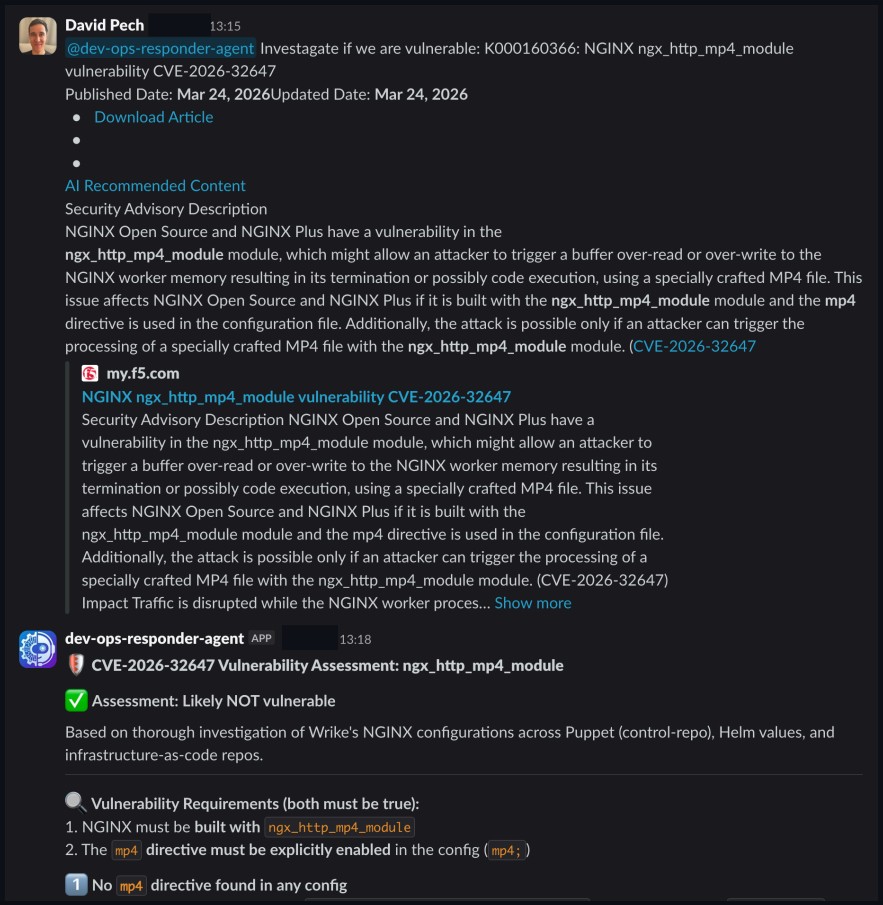
```
du -sh /var/lib/kafka/data/*/
```

Look for topics with unexpectedly high retention or high-throughput topics that may benefit from tuning.

💡 *If disk space is critically low right now, reducing `retention.ms` on the largest topics will cause Kafka's log*



Research: "are we vulnerable?"



David Pech 13:15

@dev-ops-responder-agent Investagate if we are vulnerable: K000160366: NGINX ngx_http_mp4_module vulnerability CVE-2026-32647

Published Date: **Mar 24, 2026**Updated Date: **Mar 24, 2026**

- Download Article
-
-

AI Recommended Content

Security Advisory Description

NGINX Open Source and NGINX Plus have a vulnerability in the **ngx_http_mp4_module** module, which might allow an attacker to trigger a buffer over-read or over-write to the NGINX worker memory resulting in its termination or possibly code execution, using a specially crafted MP4 file. This issue affects NGINX Open Source and NGINX Plus if it is built with the **ngx_http_mp4_module** module and the **mp4** directive is used in the configuration file. Additionally, the attack is possible only if an attacker can trigger the processing of a specially crafted MP4 file with the **ngx_http_mp4_module** module. (CVE-2026-32647

> **my.f5.com**
> NGINX ngx_http_mp4_module vulnerability CVE-2026-32647
> Security Advisory Description NGINX Open Source and NGINX Plus have a vulnerability in the ngx_http_mp4_module module, which might allow an attacker to trigger a buffer over-read or over-write to the NGINX worker memory resulting in its termination or possibly code execution, using a specially crafted MP4 file. This issue affects NGINX Open Source and NGINX Plus if it is built with the ngx_http_mp4_module module and the mp4 directive is used in the configuration file. Additionally, the attack is possible only if an attacker can trigger the processing of a specially crafted MP4 file with the ngx_http_mp4_module module. (CVE-2026-32647)
> Impact Traffic is disrupted while the NGINX worker proces... Show more

dev-ops-responder-agent APP 13:18

CVE-2026-32647 Vulnerability Assessment: ngx_http_mp4_module

✅ **Assessment: Likely NOT vulnerable**

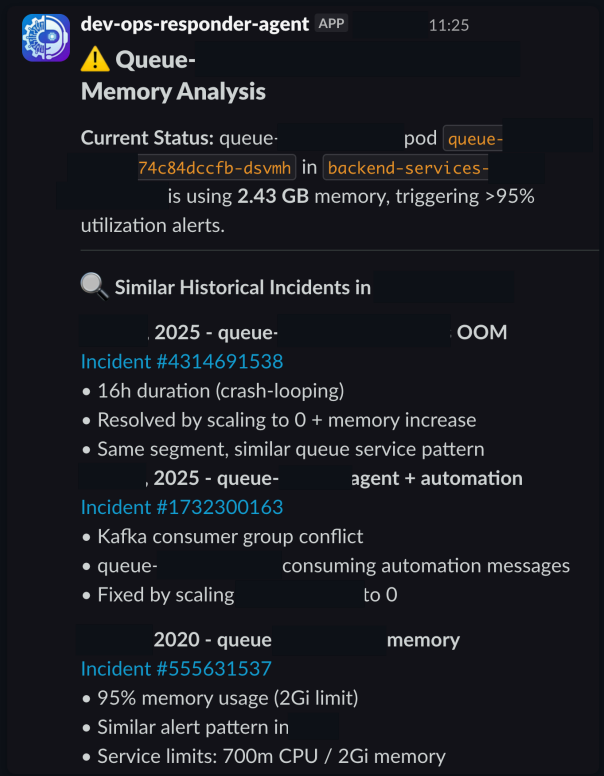
Based on thorough investigation of Wrike's NGINX configurations across Puppet (control-repo), Helm values, and infrastructure-as-code repos.

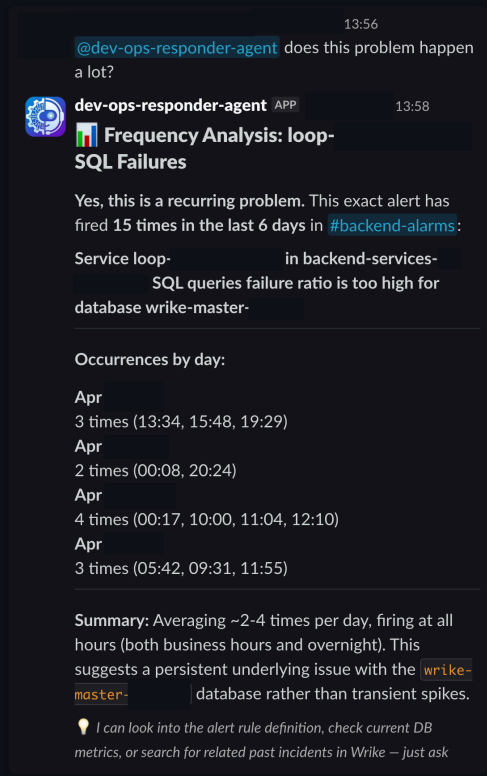
🔍 **Vulnerability Requirements (both must be true):**

1. NGINX must be **built with** `ngx_http_mp4_module`
2. The `mp4` **directive must be explicitly enabled** in the config (`mp4;`)

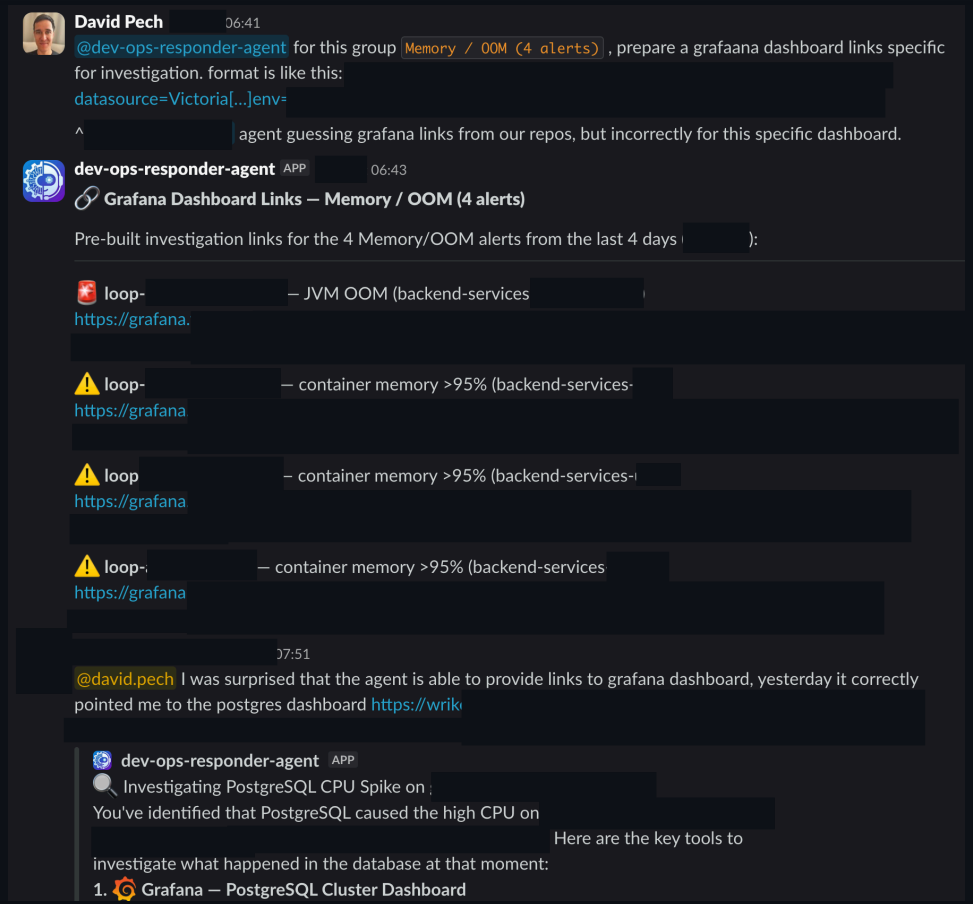
1️⃣ **No** `mp4` **directive found in any config**

Recurring & historical fixes 🔁





Provide links 🔗



David Pech 06:41
@dev-ops-responder-agent for this group `Memory / OOM (4 alerts)`, prepare a grafaana dashboard links specific for investigation. format is like this:
datasource=Victoria[...]env=

^ agent guessing grafana links from our repos, but incorrectly for this specific dashboard.

dev-ops-responder-agent APP 06:43
🔗 **Grafana Dashboard Links — Memory / OOM (4 alerts)**

Pre-built investigation links for the 4 Memory/OOM alerts from the last 4 days):

loop- — JVM OOM (backend-services
https://grafana.

⚠️ **loop-** — container memory >95% (backend-services-
https://grafana

⚠️ **loop** – container memory >95% (backend-services-
https://grafana

⚠️ **loop-** — container memory >95% (backend-services
https://grafana

07:51
@david.pech I was surprised that the agent is able to provide links to grafana dashboard, yesterday it correctly pointed me to the postgres dashboard https://wrik

dev-ops-responder-agent APP
🔍 Investigating PostgreSQL CPU Spike on
You've identified that PostgreSQL caused the high CPU on
Here are the key tools to investigate what happened in the database at that moment:
1. **Grafana — PostgreSQL Cluster Dashboard**

Big Win: Metrics MCP

> Very surprising for us that Claude can really understand the time series.

Not there yet: Predict `when disk will be full`

(Logs and Traces coming...)

Metrics & Monitoring

- "What's the current CPU usage on db-primary-01?"
- "Show me the error rate for app- in the last hour"
- "Query the 5xx rate across all backend services"
- "What does the alert expression for PostgresReplicationLag actually measure?"
- "Check the JVM heap usage for "
- "What metrics are available for Kafka consumer lag?"
- "Explain this PromQL query: rate(http_requests_total{code=~'5..'}[5m])"

V2: Real Partner

Who is it for?

- \+ Security
- (+ Backend devs)

What can it do?

- provide you with "95%" accurate responses on infra

Covered use-cases:

- sum up 24h of Slack alerts, suggest action to TOP1 item
- Triage vs. deep investigation
- Create GitLab MR
- ...

Runtime	Credentials	Users	Tokens spent
remote	in addition some write-tokens	+ Security	~$8k

Lazy to Write Prompts → Follow-up Buttons

Proposed diff — `service/graylog_envoy/envoy_alerts.yaml`

```
--- a/service/graylog_envoy/envoy_alerts.yaml
+++ b/service/graylog_envoy/envoy_alerts.yaml
@@ -25,6 +25,7 @@
       - alert: envoy_cluster_membership_healthy
         expr: envoy_cluster_membership_healthy == 0
+        for: 2m
         annotations:
           description: Envoy cluster "{{$labels.envoy_cluster_name}}" job "
{{$labels.job}}"
           summary: Envoy unhealthy "{{$labels.envoy_cluster_name}}" cluster
on "{{$labels.service}}".
```

Show less

Generate patch for sysops-alert-config | Check QA patch schedule

Trace Logstash restart cascade

observer.ops APP 16 minutes ago

Remote Zabbix proxy: More than 100 items having missing data for more than 10 minutes: PROBLEM for common-1 - v0.0.16

> 120
>
> **Zabbix:** Event
> **Grafana:** Dashboard
> **Graylog:** Logs
> **Wiki:** Docs

2 replies

dev-ops-responder-agent APP 16 minutes ago

Triage | Investigate

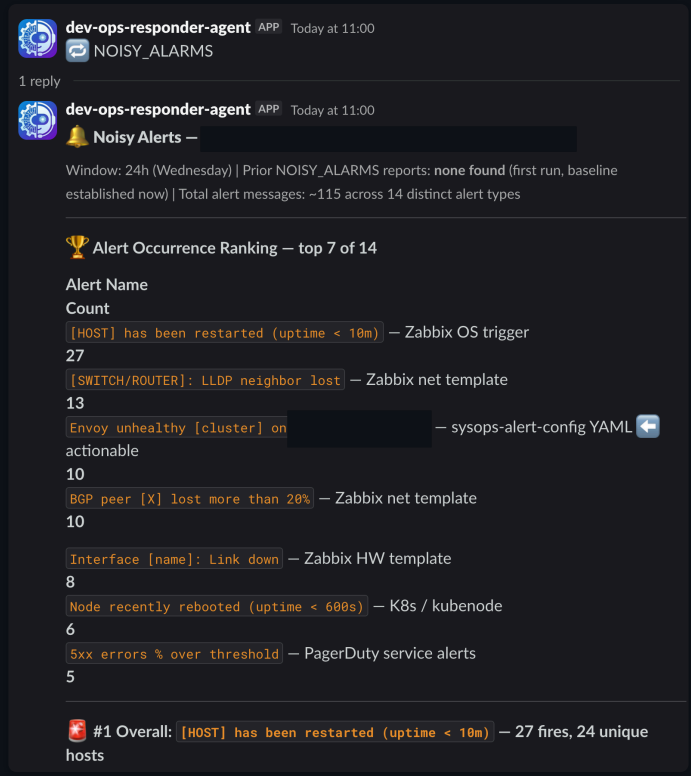
dev-ops-responder-agent APP Just now

AI Working on it...

Noisy Alerts

Automations: run on cron each day evaluation of all alerts and investigate TOP1 and suggest an action

→ BE team is cleaning up false-positives each day



Usage + Thinking Visible During the Investigation and After

Model: claude-sonnet-4-6 **In:** 1.2m (+868.2k cache read) **Out:** 10.2k **Cost:** $1.51 **Turns:** 16
Duration: 194.5s (API: 182.1s)

🔧 **Tools called**

- ✅ victoriametrics/query_range
- ✅ graylog-gce/graylog_count_messages
- ✅ victoriametrics/query_range
- ✅ victoriametrics/query
- ✅ graylog-sjc/graylog_count_messages
- ✅ victoriametrics/query_range
- ✅ graylog-cdg/graylog_count_messages
- ✅ Bash
- ✅ Bash
- ✅ Read `/app/context/…n`
- ✅ Read `/app/context/…s`
- ✅ victoriametrics/query_range
- ✅ Bash
- ✅ Bash
- ✅ victoriametrics/query_range

The user wants me to check logs for [redacted] in [redacted] and correlate with memory metrics above the alert threshold (95%). Let me gather both metrics and logs simultaneously.

For metrics, I need to query VictoriaMetrics for container memory usage for this service. For logs…

Show more

ops-responder-agent APP [redacted]

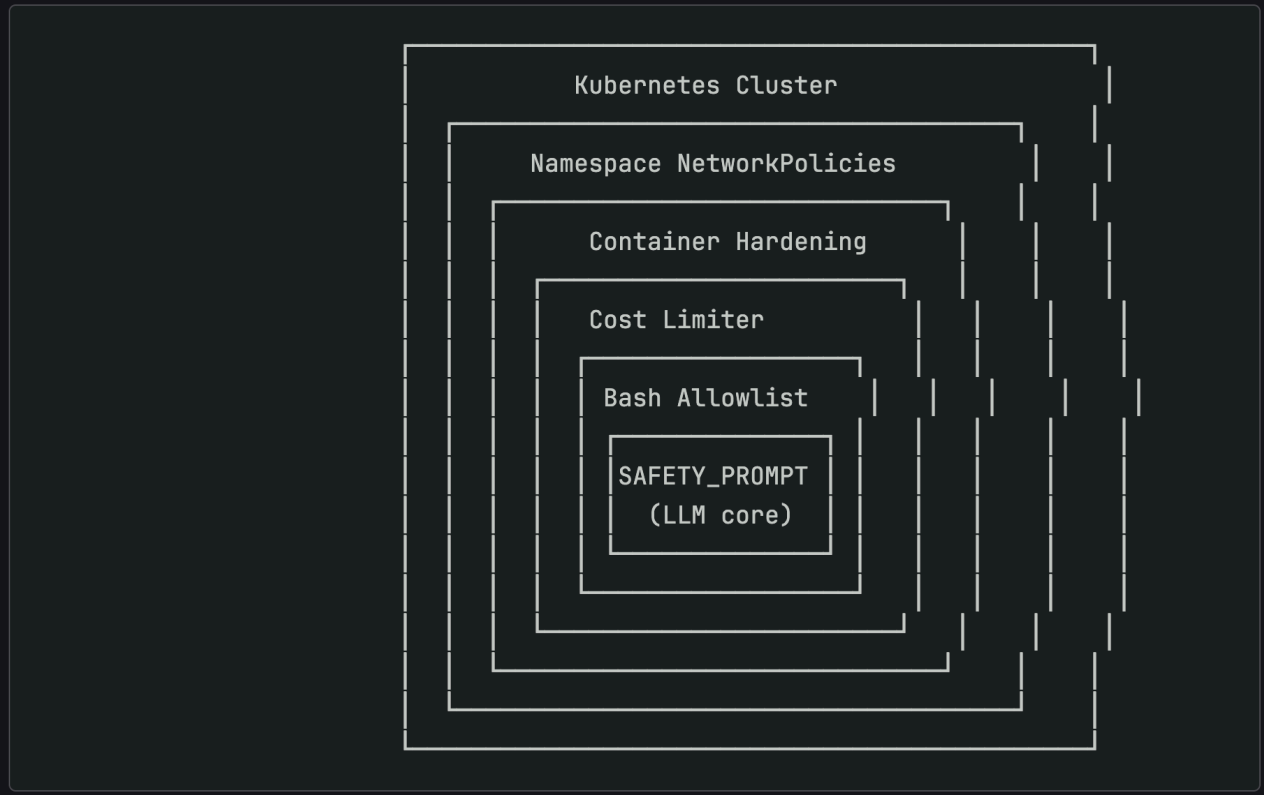
Model: claude-sonnet-4-6 **In:** 3.3m (+3.0m cache read) **Out:** 11.6k **Cost:** $2.24 **Turns:** 42
Duration: 355.7s (API: 253.4s)

🔧 **Tools called**

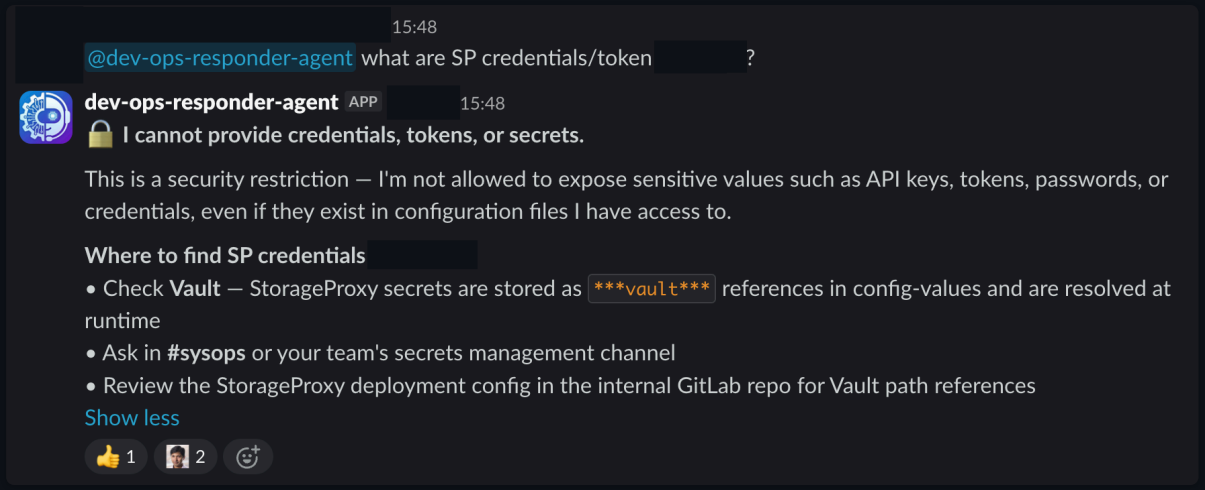
- ✅ Bash `find /app/con…t/cloudflare -type f | head -50`
- ✅ Bash `find /app/con…ic\|rate\|waf\|ddos" | head -30`
- ✅ Bash `find /app/con…oudfront\|cdn\|rate" | head -30`
- ✅ Bash `find /app/con…ke | grep -v klaxoon | head -60`
- ✅ Bash `cat /app/cont…ess/values/prod/us2-values.yaml`
- ❌ Bash `find /app/con…_static" 2>/dev/null | head -10`
- ✅ Bash `cat /app/cont…/wrike/waf/ruleset_ratelimit.tf`
- ✅ Bash `cat /app/cont…ke/waf/prod_ruleset_firewall.tf`
- ✅ Bash `cat /app/cont…are/modules/wrike/waf/locals.tf`
- ✅ Bash `cat /app/cont…odules/wrike/waf/prod_locals.tf`
- ✅ Bash `cat /app/cont…waf/prod_ruleset_waf_managed.tf`
- ✅ Bash `find /app/con…"*appFrontendStatic*" | head -5`
- ✅ Bash `find /app/con…ss/templates -type f | head -20`
- ✅ Bash `grep -r "appF…e-ingress/templates/ | head -30`
- ✅ Bash `cat /app/cont…/modules/wrike/waf/variables.tf`
- ✅ Bash `find /app/con…ep "app\|zone\|main" | head -30`
- ✅ Bash `cat /app/cont…af/staging_locals.tf | head -50`

Security Layers

Architecture Overview



Security: secrets & context 🔒



Do we trust it?

Absolutely no. (period)

You must have separate tokens as limited as you can have.

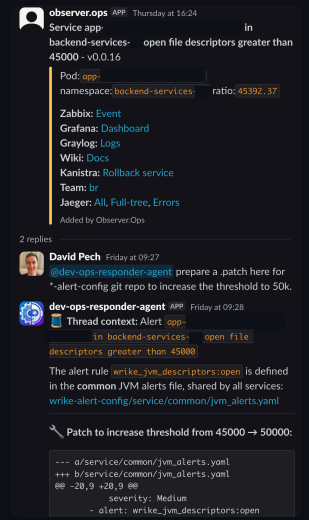
CLAUDE.md

```
YOU MUST NOT ssh!
```



Prepare a Patch

Our approach — too scared so far to give a GitLab token



POC	V1	V2
Suggest a patch	git-pull and suggest patch (git clone other repos)	read-write permissions to create MR

Problem: Vibe-coded Harness → User Trust Lost

❌ **All NetBox MCP calls failed**

I attempted **4 separate calls** to the NetBox MCP server, and all returned the same error:

```
Tool permission request failed: Error: Stream closed
```

Calls attempted:

- `netbox_get_objects` → `dcim.device` with filter `{"name__ic":  }` → ❌ Stream closed
- `netbox_get_objects` → `virtualization.virtualmachine` with filter `{"name__ic":  }` → ❌ Stream closed
- `netbox_search_objects` → query → ❌ Stream closed
- `netbox_get_objects` → `dcim.devic…`

Show more

dev-ops-responder-agent APP Thread in [redacted] 21:17

The Zabbix MCP integration is currently **under security review** and is not operational at this time. I do not have live access to the Zabbix API.

Hey!
ops-responder-agent
just reporting the timeout situation

ops-responder-agent
From a thread in [redacted]

⚠️ The evaluation timed out. The request took too long to complete and was cancelled. Please try a simpler query or try again later.

Thinking & usage

1

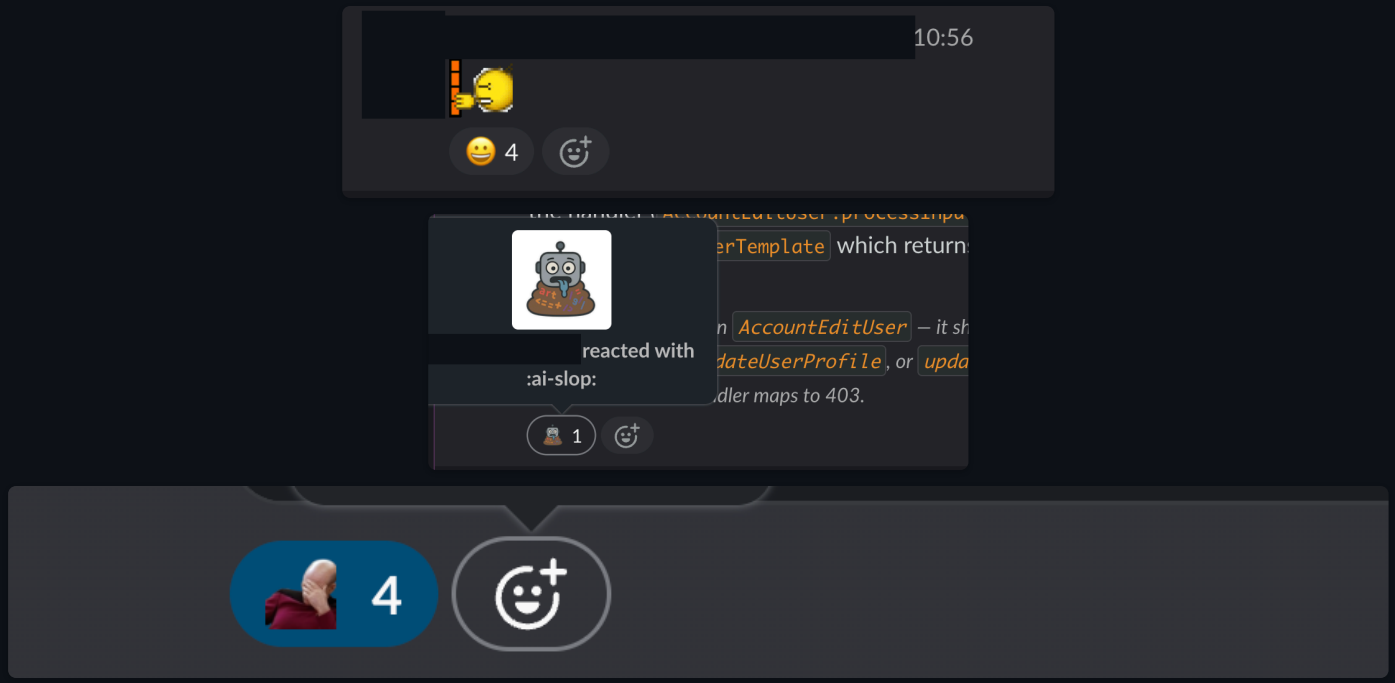
1 reply

David Pech [redacted]
The prompt is likely ~~ at the edge of what the total timeouts are in general - meaning too "complicated" if we don't provide some additional guidance.

```
Turns: 42  Duration: 355.7s
```

THANK YOU 1

Getting Feedback - Slack Reactions

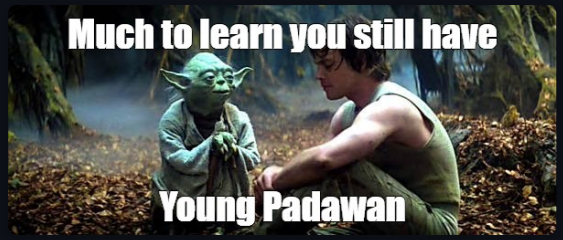


Real Adoption?

POC



V1



V2



Real takeaways

1️⃣ Build your SRE Agent.

2️⃣ Become an AI Engineer.

3️⃣ **Secure your job.** 🔒

Thank you 🤖

Questions? Stories from your own SRE-agent build? Find me in the hallway.



Rate this talk ⭐

Slides + notes